



Hemlata Gautam
Bachelor of Technology
in Computer Science Engineering
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Github | Linkedin
Codeforces | Portfolio

CAREER OBJECTIVE

My aim is to become a skilled and adaptable Software Engineer with a strong background in AI/ML and software development. I am passionate about exploring new technologies, committed to delivering quality work. My fluency in English, Hindi, and Urdu helps me communicate ideas clearly and work efficiently in multicultural environments.

EDUCATION

Degree	Institute/Board	cGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	6.54	2022-2026
Senior Secondary	Central Board of Secondary Education	92.8%	2021
Secondary	Central Board of Secondary Education	94.8%	2019

EXPERIENCE

- **Summer Internship - Cloud-based AC temperature control IoT application**
Canada Project | ESP32, MQTT(Mosquitto), Fast API, Flutter, Postgresql

May 2025 - July 2025
Offer Letter

– Architected a cloud-based IR-AC control system using an ESP32 microcontroller, Mosquitto MQTT broker and microservices backed by PostgreSQL to enable reliable remote temperature management.

– Developed a Flutter app for real-time temperature control, telemetry, and scheduling.
- **ML Internship - IEEE SPS VIP Cup Winner’24: Retinal OCT Image Classification**
Under the supervision of Dr. Puneet Goyal, IIT Ropar | Python, Tensorflow, OpenCV

May 2024 - July 2024
Github

– Secured **1st position globally** to qualify for the finals at the ICIP Conference in Abu Dhabi.

– Paper produced - "Classification of Noisy Retinal OCT Images into Healthy, DME and Non-Diabetic Disorders"

– **Tasks Performed:**
1. Denoising, 2. Super Resolution, 3. Classification

– **Models Implemented** - Self2Self Net, cGAN, CycleGAN, SRGAN, 2Encoder-1Decoder ESRGAN, ResNet50, EfficientNetB0 based CNN architectures
- PROJECTS
- **CogniGrade App : Automatic Answerscript Checker**
DEP | Gemini API,Python Fast API, HTML, CSS, JS, Postgresql

March 2025 - May 2025
Demo

– It is a web application to automate the evaluation of student’s answer scripts.

– Utilized the Gemini API for both handwritten text extraction and grading based on content relevance.

– Optimized the assessment process, significantly reducing manual evaluation effort for educators.

• **ADOBE : Image Classification and Artifact Detection**
Machine Learning , XAI | VLM, Hugging Face, ESRGAN, Grad-CAM

Dec 2024- Dec 2024
Github

– Achieved 98% accuracy on CIFAKE dataset of size 32x32 images under FGSM attacks with 150 ms CPU inference.

– Localized regions of interest using Grad-CAM and employed a Vision-Language Models (VLM) to detect artifacts.

Secured 11th position among 28 Teams in IIT Bombay.

• **Self Balancing Robot**
Tinkering Lab | Arduino IDE, HC05 Bluetooth APK

Oct 2023 - Dec 2023
Github

– Constructed a Self Balancing Robot that balances on 2 wheels using gyroscopic sensors and an accelerometer.

– Utilized PID (Proportional-Integral-Derivative) control for accurate movement, enabling the robot to be operated via Bluetooth from a smartphone, thereby improving user interaction and control precision.

• **Interactive Web Games Creation**
Personal Project | JavaScript, HTML5, CSS3, Postgres SQL, DBeaver

Jul 2024 - Jul 2024
Github

– Developed a collection of engaging web games, including Parking Zone, Flappy Bird, and Bouncing Ball

– Integrated score tracking and leaderboard features to enhance competitive gameplay and user retention.

• **Minor Projects:**

Sep 2023 - Apr 2024
Github

– Efficient Spell Checker, RISC-V Assembler and CUCU Compiler.

TECHNICAL SKILLS

• **Programming Languages:** C/C++, Python, JavaScript, HTML5, CSS, PL/pgSQL, Java, Risc-V Assembly, Dart

• **Libraries & Platforms:** Tensorflow - Keras, PostreSQL, DBeaver, PIL, Numpy, Matplotlib, Kaggle, Github, Overleaf , Linux

ACHIEVEMENTS

• **IEEE VIP Cup Winner’24**, Secured **1st position** globally at the ICIP Conference in Abu Dhabi.

2024

• **Pupil at Codeforces**, Rating Achieved **1244**

2024

• **Silver Medal**, Relay Race, Aarohan in IIT Ropar

2023

• **8th Rank in Inter IIT**, Secure position in Top 10 among 60 students in Canvas Painting, IIT Kharagpur

2023

POSITIONS OF RESPONSIBILITY

– **Coordinator**, Art Club, IIT Ropar

May. 2023 - May. 2024

– **Co-Head** , Zeitgeist (Annual Fest), IIT Ropar

Oct 2023- Nov 2023

– **Event Co-Head**, Art Club in Zeitgeist, IIT Ropar

Oct 2023- Nov 2023

KEY COURSES TAKEN

– **Machine Learning Specialisation:** Stanford, Andrew Ng. - Certificate

– **CSE:** DSA, Discrete Maths, Programming Paradigms, Computer Architecture, TOC, Computer Networks, Artificial Intelligence